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Quantifying the Impact of Elon Musk's Digital Footprint on TSLA Returns Using Wikipedia

In the market-place, one convinces with gestures.
But reasons make the populace distrustful.

— Friedrich Nietzsche, Thus Spoke Zarathustra

Abstract

Financial markets often react sharply to the public behavior of high-profile executives, yet such attention is difficult to measure in a timely and structured way. This paper studies whether any Wikipedia activity, such as editing rate, daily views, etc. can serve as a real-time or leading proxy for CEO-related attention in the case of Tesla and Elon Musk. We examine the editing histories of the Elon Musk and Tesla, Inc. pages and relate them to Tesla's daily stock returns while controlling for broader market movements and Elon Musk's online activity on X (formerly Twitter). In particular, our results show that editing activity on the Elon Musk Wikipedia page is statistically significant in explaining Tesla's returns, with one additional edit on the previous trading day being associated with an average decrease of approximately 0.03% in today's TSLA returns.

1 Introduction

Tesla provides an especially revealing setting for studying attention-driven market dynamics because the company's public image is unusually intertwined with that of its chief executive, Elon Reeve Musk. Over the past several years, Musk's statements, interviews, and online presence have repeatedly coincided with sharp movements in Tesla's stock price, suggesting that CEO visibility can itself become economically relevant. One widely cited example occurred on May 1, 2020, when Musk tweeted that "Tesla stock price is too high imo," after which Tesla shares fell by roughly 10% over the trading day. This makes Tesla a useful case for examining whether public attention directed at an individual executive can be separated from attention directed at the firm. At the same time, such attention is difficult to quantify in a timely and systematic way, since the underlying signals are dispersed across heterogeneous sources such as social media, news coverage, interviews, and public appearances.

In this setting, Wikipedia may provide a useful intermediate signal of public attention. Editing activity is generated by a relatively engaged group of users who actively monitor, revise, and update information in response to recent events. Moreover, most Wikipedia articles are not under "Pending changes" protection, allowing editors to submit their changes with immediate approval. Although Wikipedia is not a direct measure of investor sentiment, it can reflect moments when public interest becomes sufficiently intense to trigger collective information production. This idea is consistent with prior research showing that Wikipedia activity can help predict real-world outcomes. For example, Mesty'an, Yasseri, and Kert'esz show that activity surrounding movie-related Wikipedia pages can predict box-office success before release [1]. More critically for financial applications, Moat et al. demonstrated that changes in how frequently Wikipedia users access and edit financially related pages can serve as an early indicator of stock market movements [2], establishing Wikipedia as a valid lens for investor attention.

Building on this intuition, this paper examines whether activity on Wikipedia can serve as a real-time proxy for investor attention in Tesla's stock performance. We focus on two articles that are especially relevant to the company's public image and market dynamics: **Elon Musk** [3] and **Tesla, Inc.** [4].

The setting motivates several central research questions:

- **RQ1:** Does any activity on the Elon Musk page correspond to lower Tesla stock returns?
- **RQ2:** If yes, what is more influential: edits or views?
- **RQ3:** What has larger association with Tesla's returns: activity on the Elon Musk page or activity on the Tesla, Inc. page?

2 Data collection & description

The empirical analysis combines Wikipedia edit histories, Wikipedia daily pageviews, controls like Elon Musk's activity on X (Twitter), and daily SP500 log return. On the Wikipedia side, our main contribution is a Python-based API that retrieves the full revision metadata for the *Elon Musk* [3] and *Tesla, Inc.* [4] articles and stores the output in a structured dataset. The raw revision fields returned by this API are summarized in Table 1.

On the financial side, we collected daily time-series data for Tesla stock (TSLA) and for the S&P 500 index using *yfinance*, from which we constructed Tesla and market log returns,

$$log_returns_t := \log\left(\frac{close_t}{close_{t-1}}\right),$$

and

$$mkt_returns_t := \log\left(\frac{spclose_t}{spclose_{t-1}}\right).$$

To align all count-based signals with market outcomes, we define trading day as

$$\mathscr{T}_t := [16{:}00 \text{ ET on day } t-1,\ 16{:}00 \text{ ET on day } t],$$

and resample count variables to this interval. This applies to the Wikipedia edit counts and to the X activity count msk_tweets_t constructed from the Kaggle dataset *All Elon Musk's Tweets* [5]. From the revision histories, we then engineer the article-level variables msk_edits_t, $tsla_edits_t$, $msk_unique_editors_t$ and $tsla_unique_editors_t$. By contrast, the pageview variables msk_views_t and $tsla_views_t$ remain at the calendar-day frequency because Wikipedia provides them only in daily aggregated form. Weekend observations are removed because U.S. equity markets are closed. Table 2 summarizes the main dataset used for analysis.

Table 1: Custom API output table: revision-level Wikipedia metadata

Variable	Description	Type
`revid`	Unique identifier of the revision	Integer
`timestamp`	Date and time at which the revision was recorded	Datetime
`user`	Username or IP-based identifier of the editor	Text
`size`	Page size after the revision	Integer
`sha1`	Content hash used to track repeated article states	Text
`flags`	Revision flags returned by the Wikipedia API	Text
`intro_text`	Introductory article text associated with the revision snapshot	Text
`minor`	Indicator for whether the revision was marked as minor	Binary

Table 2: Description of the main time series dataset used for analysis

Variable	Description	Frequency
$log_returns_t$	Tesla daily log return computed from consecutive closing prices	Daily
$mkt_returns_t$	S&P 500 daily log return computed from consecutive index closes	Daily
mkt_volume_t	S&P 500 trading volume	Daily
$tsla_volume_t$	Tesla trading volume	Daily
msk_edits_t	Total edits on the *Elon Musk* article over trading day t	Daily
$msk_unique_editors_t$	Number of distinct editors active on the *Elon Musk* article over trading day t	Daily
$tsla_edits_t$	Total edits on the *Tesla, Inc.* article over trading day t	Daily
$tsla_unique_editors_t$	Number of distinct editors active on the *Tesla, Inc.* article over trading day t	Daily
msk_tweets_t	Number of Elon Musk posts on X assigned to trading day t	Daily
msk_views_t	Daily Wikipedia pageviews for the *Elon Musk* article (calendar-day series)	Daily
$tsla_views_t$	Daily Wikipedia pageviews for the *Tesla, Inc.* article (calendar-day series)	Daily

3 Data Analysis

The initial plot of daily edit frequency (Figure 1) of both articles over log of TSLA daily returns only motivates the first research question discussed before, as highly-edited periods often coincide with volatile stock market time windows.

3.1 Stationarity of TSLA log-returns

Moreover, the plot of log returns visually suggests *stationarity*, which, if formally confirmed, supports treating the series as suitable for time-series analysis without additional transformation. To assess this property more formally, we applied the Augmented Dickey–Fuller (ADF) test [6] to the Tesla return series. Also, the period after 2020 (inclusive) seems to be the most "interesting" in a sense of volatility of Wikipedia editing rate for Elon Musk. Moreover, this period is distinguishable for his persona because of Twitter aquisition and the increase in media presence. That is why, for further research we will fix this time period.

The ADF test results may be accessed in Table on the right. With the p-value of 0.0000 we can confidently reject the null hypothesis and confirm stationarity, which is really expected for log returns in stock markets. *Note: the stationarity is also confirmed for shortened time period after 2020.*

Statistic	Value
ADF test statistic	-63.2207
p-value	0.0000
Number of lags used	0
Number of observations	3955
1% critical value	-3.4320
5% critical value	-2.8623
10% critical value	-2.5672
Information criterion (IC best)	-14984.0330

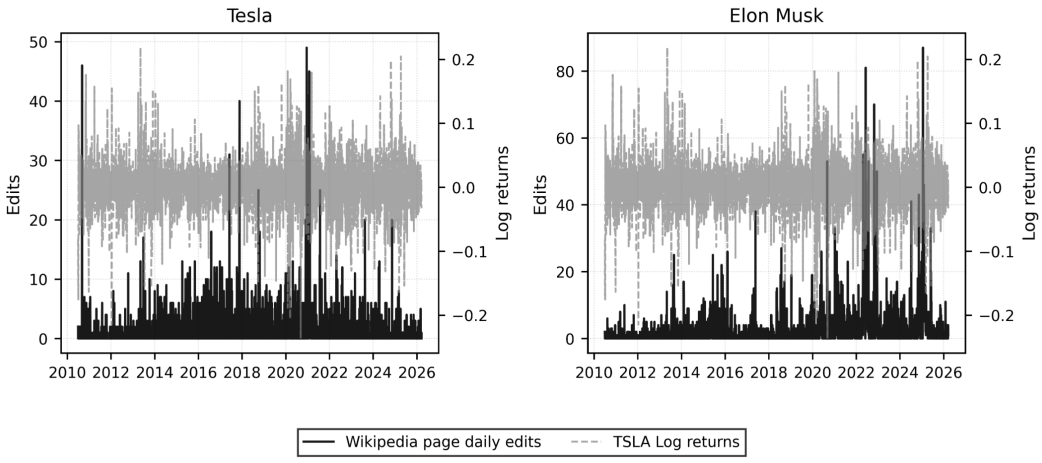

Figure 1: Daily edit frequency for the Elon Musk and Tesla, Inc. Wikipedia pages and TSLA daily log returns.

3.2 Multicollinearity

Next, it is worth considering correlations between predictors to check for multicollinearity and choose a decent set of predictors. Figure 2.1 reports the correlation matrix before excluding variables or introducing transformed ones, where Figure 2.2 summarizes the set after this step. From Figure 2.1, we can see that *msk_edits* is highly correlated with *msk_unique_editors* and that *tsla_edits* is highly correlated with *tsla_unique_editors*. While the unique-editor variables could serve as an interesting source of information, we decided to exclude them and let the coefficients on the edit-count variables absorb that effect because these are more intuitive to interpret. Moreover, $msk_wiki_views_t$ is highly correlated with $tsla_wiki_views$. To exclude choosing between them both, it was decided to take only unexplained part of $tsla_wiki_views_t$ (define $tsla_views_unex_t$ as residuals from estimating $msk_views = \gamma_0 + \gamma_1 tsla_views + u_t$). Pleasantly, no other statistics extracted from Wikipedia pages seem to be strongly correlated across the two articles, allowing us later to include a mix of them in a single regression.

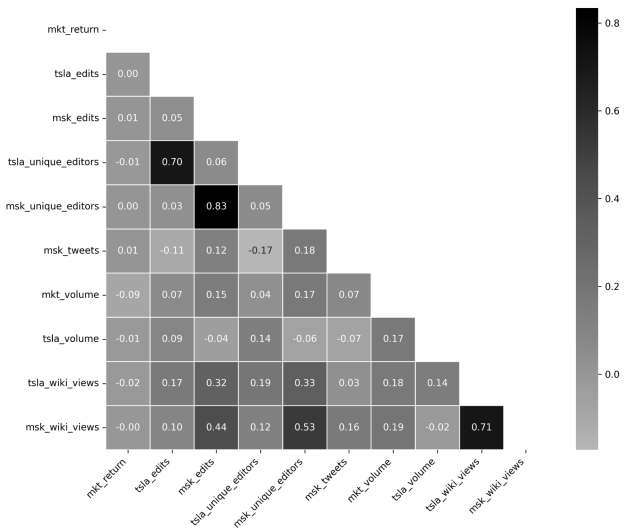

Figure 2.1: Correlation matrix before excluding variables and introducing transformed predictors.

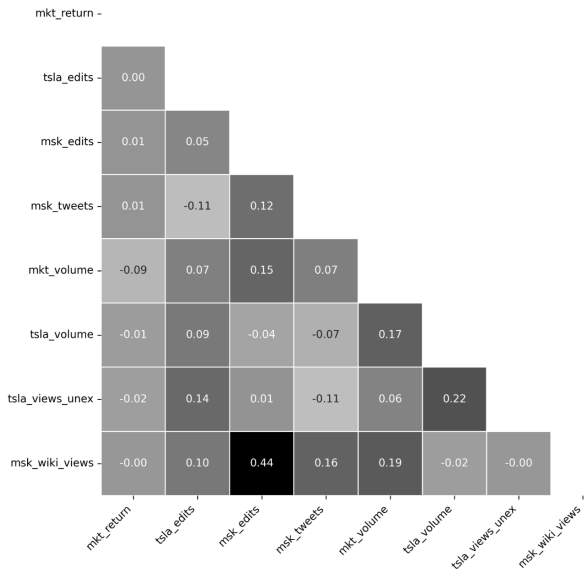

Figure 2.2: Correlation matrix after transformations.

3.3 Lags of variables

Before building the final model, we estimated regressions of daily log returns on lagged values of each predictor up to five days. This was done to understand whether past values of the variables contain useful information for predicting returns and to help guide the choice of lags in the final specification. Market returns were included as a control variable but not shown in the heatmap.

The *results* 4 show that most lagged effects are very small. We observe a small negative effect at lag 1 for $msk_edits_t(-0.0005)$ and a similar magnitude effect at lag 2 for $tsla_edits_t$. However, the distinguishable effect for $tsla_edits_t$ only at the second lag, together with a complete absence of effects at lags 0 and 1, is more likely a sign of randomness rather than a true delayed relationship, considering the fragile nature of daily log returns variable. For this reason, we will only include a lagged version of msk_edits_t in final specifications. *(Lags of mkt_return_t were estimated separately to make the diagram readable since the magnitude of coefficient is much larger, which would gather all color intensity. However, no lags after 0 of it were statistically significant.)*

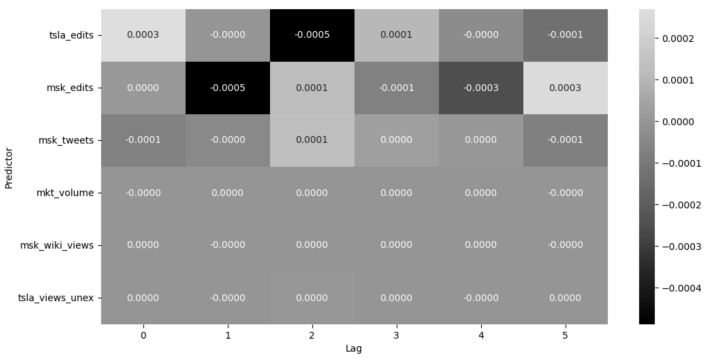

Figure 4: Finite Distributed Lag Model Coefficients for each predictor

4 Methods

4.1 Model specification

Based on Section 3, we estimate a linear model in a specification including three groups of regressors: general market controls, Tesla-specific attention variables, and Elon Musk-specific attention variables. Market return is included to control for broad movements in equity markets, number of tweets attemp to isolate at least some CEO-related dynamics while log-transformed market and Tesla trading volumes are used to reduce the influence of extreme observations.

$$\begin{aligned} log_return_t = \alpha_0 + \alpha_1 mkt_return_t + \alpha_2 \log(mkt_volume_t) + \beta_1 \log(tsla_volume_t) + \beta_2 tsla_edits_t \\ + \beta_3 tsla_views_unex_t + \gamma_1 msk_tweets_t + \gamma_2 msk_edits_t + \gamma_3 msk_edits_{t-1} + \gamma_4 msk_views_t + u_t. \end{aligned} \tag{1}$$

4.2 Gauss-Markov assumptions assessment

Under **TS.1**, linearity is ensured by the specification, while weak dependence is supported through stationarity analysis (Section 3.1).

TS.2 requires the absence of perfect collinearity. This was assessed using the correlation matrix in Figure 2.2, with highly correlated variables removed or treated.

TS.3 assumes contemporaneous exogeneity. While this cannot be tested directly, we improve its plausibility by including control variables such as mkt_return_t and msk_tweets_t. At the same time, the Wikipedia activity variables are used as proxies in this research, so some violation of this assumption is intentionally allowed and is recognized as a limitation.

To test for homoskedasticity (**TS.4**), we use *Breusch-Pagan* test, which regresses estimated errors $\hat{u}_t$ against independed variables. The results (3) $(p \approx 0)$ suggest strong evidence of *heteroskedasticity*. To account for this, the model will be estimated using HC1 robust standard errors.

Serial correlation (**TS.5**) is accessed in the model output using *Ljung-Box* test [7]. When tested (3) on 10 lags, no significant evidence of serial correlation appeared.

Together, these assessments provide the basis for reliable hypothesis testing under the asymptotic time-series regression framework.

Test	**Stat. 1**	p**-value 1**	**Stat. 2**	p**-value 2**
Breusch–Pagan	$LM = 221.7103$	9.1415×10^{-43}	$F = 29.3573$	6.4746×10^{-47}
Ljung–Box (10 lags)	$Q = 13.3857$	0.2029	—	—

Table 3: Residuals diagnostics tests results for model 1

4.3 Hypotheses

To evaluate the research questions, we test multiple linear restrictions on the coefficients using F-tests and individual coefficient restrictions (or linear combinations) using Wald t-tests [8].

RQ1 asks whether Musk-related Wikipedia variables have explanatory power for Tesla returns after controlling for market, Twitter activity of Musk and Tesla-specific factors. We therefore test

$$H_0 : \gamma_2 = \gamma_3 = \gamma_4 = 0 \qquad \text{vs.} \qquad H_1 : \text{at least one } \gamma_j \neq 0.$$

RQ2 asks whether one edit is indeed a more valuable unit of participation than one view (lagged views were added to specification. For this test, the model is defined as (1) $+\gamma_5 msk_wiki_views_{t-1}$):

$$H_0 : \gamma_2 + \gamma_3 = \gamma_4 + \gamma_5 \qquad \text{vs.} \qquad H_1 : \gamma_2 + \gamma_3 > \gamma_4 + \gamma_5$$

RQ3 compares CEO-related attention with firm-related attention. We compare the effect of Wikipedia article on Tesla, Inc with the total effect of Musk's article. Only for this test, $\beta_4 tsla_edits_{t-1}$ was added to (1) to completely match the included variables from Musk article:

$$H_0 : \sum_{k=2}^{4} \beta_k = \sum_{k=2}^{4} \gamma_k \qquad \text{vs.} \qquad H_1 : \sum_{k=2}^{4} \beta_k < \sum_{k=2}^{4} \gamma_k.$$

5 Results

5.1 Regression summary and test results

The estimated regression summary may be accessed in Table 4. While most estimates in initial specification did not show statistical significance or were designed as control (market return), msk_edits_{t-1} posseses negative estimate -0.0003 with 95% confidence, which gives editing activity of Elon Musk article a small yet statistically significant predictive power. This motivated an additional question **RQ0**, whether the aggregate effect after of edits after lags 0 and 1 is also significant. ($H_0 : \theta := \gamma_2 + \gamma_3 = 0$).

The hypothesis-test outcomes are summarized in Table 5. Together, they show that the intermediate edit effect and the joint Musk-article effect are statistically significant, while the comparison with Tesla-page activity is only marginally significant in the one-sided RQ3 specification. Basically, we **reject** all four null hypotheses corresponding to **RQ0-RQ3**.

Variable	**Coef.**	**Std.Err.**	**z**	**P>|z|**	**[0.025**	**0.975]**
Intercept	-0.1416	0.0606	-2.336	0.020	-0.261	-0.023
mkt_return	1.6559	0.1105	14.990	0.000	1.439	1.872
log(mkt_volume)	-0.0011	0.0012	-0.901	0.367	-0.003	0.001
log(tsla_volume)	0.0092	0.0029	3.128	0.002	0.003	0.015
tsla_edits	0.000098	0.00024	0.407	0.684	-0.000	0.001
tsla_views_unex	-2.08e-07	4.33e-07	-0.480	0.631	-1.06e-06	6.42e-07
msk_tweets	-4.24e-05	4.23e-05	-1.001	0.317	-0.000	4.06e-05
msk_edits	-0.0001	0.00011	-0.944	0.345	-0.000	0.000
msk_edits_{t-1}	-0.0003	0.00018	-1.711	0.087	-0.001	4.45e-05
msk_wiki_views	1.50e-09	1.97e-08	0.076	0.939	-3.72e-08	4.02e-08
Model statistics						
Observations						1326
R-squared						0.305
Adj. R-squared						0.300
F-statistic						27.03
Prob (F-statistic)						3.34e-43
AIC						-5052
BIC						-5000
Durbin-Watson						2.046

Table 4: OLS Regression Results

Question	**Test**	**Reported result**	***p*-value**	**Statistic**	**Conclusion**
RQ0	Wald t-test	$c_0 = -0.0004$, 95% CI $[-0.001, -9.19 \times 10^{-5}]$	0.011	$z = -2.542$	Reject H_0 at 5%
RQ1	F-test	Joint test of $\gamma_2 = \gamma_3 = \gamma_4 = 0$	0.0312	$F = 2.963$	Reject H_0 at 5%
RQ2	Wald t-test	$c_0 = -0.0004$, 95% CI $[-0.001, -9.18 \times 10^{-5}]$	0.011	$z = -2.559$	Reject H_0 at 5%
RQ3	Wald t-test	$c_0 = -0.0005$, 95% CI $[-0.001, 6.73 \times 10^{-5}]$	0.084	$z = -1.728$	Reject H_0 only one-sided

Table 5: Summary of intermediate and research-question test results

5.2 Interpretation

The results for $RQ1$ indicate that Wikipedia activity related to Elon Musk's article is jointly relevant for explaining Tesla returns. Specifically, the joint significance test of $H_0 : \gamma_2 = \gamma_3 = \gamma_4 = 0$ is rejected ($F = 2.963$, $p = 0.0312$), which implies that the Musk-related Wikipedia regressors—current edits, lagged edits, and pageviews—have a statistically significant combined effect on TSLA daily returns. Also, the strongest signal comes from editing activity. In particular, the aggregate effect of msk_edits_t and msk_edits_{t-1}, that is, the linear combination $\gamma_2 + \gamma_3$, is statistically significant even at the 98% confidence level ($z = -2.542$, $p = 0.011$). On average, c.p, one additional edit on the Elon Musk Wikipedia page across the current and previous trading day is associated with an approximately $\hat{\gamma}_2 + \hat{\gamma}_3 = 0.04\%$ decrease in today's TSLA return. Moreover, msk_edits_{t-1} is separately significant at the 95% level in the one-sided test (one-sided $p = 0.0435$), showing that, one additional edit on Musk's page yesterday is associated with roughly a 0.03% average decrease in today's TSLA return, holding general stock market dynamics and Musk's activity on twitter constant.

The answer to $RQ2$ confirms the initial intuition that one edit is a much stronger unit of active participation than one pageview: the edit coefficients are economically larger, while pageviews are not separately significant, with *msk_wiki_views* itself insignificant ($p = 0.939$).

Finally, $RQ3$ is supported in the expected direction. The estimated difference between Musk-page and Tesla-page effects has the anticipated sign and is statistically significant under the pre-specified one-sided hypothesis (z = -1.728, p = 0.084, one-sided $p/2 < 0.05$). Taken together, this provides evidence that CEO-related Wikipedia activity is more strongly associated with Tesla returns than activity on the firm's own page.

The overall goodness of fit is moderate for a daily-return model. The estimated $R^2 = 0.305$ and adjusted $R^2 = 0.300$ imply that approximately 30% of the variation in Tesla's daily log returns is explained by the included regressors. For financial return data, where noise is typically large, this represents a meaningful explanatory contribution but also indicates that about 70% of return variation still remains outside the model.

6 Limitations and Future Directions

While this project successfully utilizes Wikipedia editing activity as a proxy for CEO-driven market shocks, several structural constraints must be acknowledged

- The research is limited by the costs of relevant data. Highly valuable across traders, historical access to the official Twitter/X API and comprehensive news aggregation services is restricted by expensive commercial tiers.
- Hence, since we utilize openly available dataset of Musk's posts on Twitter, which may be edited by author before posting, we cannot fully trust it.
- Historical evidence (as presented in introduction) suggests that indeed online posts of Elon Musk should be significant in explaining TSLA returns. Hovewer, we use the overall daily tweets as a metric, which assumes that every post is meaningful. Conversely, with an average of 25 posts a day after 2020, most posts do not posses much informative value.

Therefore, further extensions of this research should:

- 1. Verify the validity of dataset with Musk's tweets or simply gather own verified source using official API access
- 2. Apply Natural Language Processing (NLP) methods to somehow classify the informational value of each post of Tesla's CEO.
- 3. Buy access to available online services providing metric like count of mentions in news/articles of Elon Musk persona or Tesla as a brand
- 3*. Build own API fetching numerous media resources and collect the data over sufficiently long time-period.

These extensions would allow us to isolate msk_edits effect even more, creating a much more rigorous research about Musk's digital footprint quantification, which may even cause edits to lose statistical significance. But at current stage, it effectively works as an available and free proxy variable, gathering effect of unavailable or costly sources.

7 Conclusions

This project set out to examine whether Wikipedia activity can serve as a real-time or leading proxy for investor attention and whether such attention is meaningfully linked to Tesla's stock returns, particularly through the lens of Elon Musk's public visibility.

First, the findings show that Wikipedia activity related to Elon Musk is jointly significant in explaining Tesla returns. This supports the idea that bursts of attention around the CEO are systematically associated with market movements. The strongest and most consistent signal comes from editing activity rather than passive pageviews, indicating that active information production is a more informative proxy for attention than simple consumption.

Second, the analysis confirms intuition that a single edit carries substantially more informational content than a single view, while pageviews are not statistically meaningful once controls are included. This highlights the importance of distinguishing between different forms of online attention,

Third, comparing the two pages shows that Musk-related Wikipedia activity has a stronger association with Tesla returns than activity on the Tesla, Inc. page itself. This suggests that for some firms, in our case, Tesla, Inc. attention directed toward the CEO plays a more central role in explaining short-term stock market dynamics than attention directed at the firm.

Overall, the model explains a moderate share of daily return variation, which is substantial in the context of financial return data, but also confirms that most movements remain driven by unobserved factors.

Taken together, the results support the main aim of the study: Wikipedia activity, particularly edits to the Elon Musk page, captures economically relevant attention shocks that are systematically related to Tesla's stock returns. Moreover, this paper may be useful to traders or co-owners, as it points out a signal that may be much more important than the brand quality itself: the intensity and dynamics of public attention directed at the firm's key decision-maker, which can act as an early indicator of market-sensitive information flows.

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Link to GitHub repository with code, data used, and the complete, explained & reproducible workflow of fetching the data and conducting the analysis: `https://github.com/ruzhylopv/wikipedia-proxy-research$0`